

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Medium

Question

$x^2 + 58x + y^2 = 0$

In the xy -plane, the graph of the given equation is a circle. What are the coordinates (x, y) of the center of the circle?

Answer

- A. $(0, 29)$
- B. $(0, -29)$
- C. $(29, 0)$
- D. $(-29, 0)$

Correct Answer: D

Rationale

Choice D is correct. It’s given that in the xy -plane, the graph of $x^2 + 58x + y^2 = 0$ is a circle. The equation of a circle in the xy -plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where the coordinates of the center of the circle are (h, k) and the radius of the circle is r . By completing the square, the equation $x^2 + 58x + y^2 = 0$ can be rewritten as $\left(x^2 + 58x + \left(\frac{58}{2}\right)^2\right) + y^2 = 0 + \left(\frac{58}{2}\right)^2$, or $(x^2 + 58x + 841) + y^2 = 841$. This equation is equivalent to $(x + 29)^2 + y^2 = 841$, or $(x - (-29))^2 + (y - 0)^2 = 841$. Therefore, h is -29 and k is 0 , and the coordinates (x, y) of the center of the circle are $(-29, 0)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.